Paper 27

HEAT TRANSFER TO STEAM–WATER MIXTURES FLOWING IN UNIFORMLY HEATED TUBES IN WHICH THE CRITICAL HEAT FLUX HAS BEEN EXCEEDED

By A. W. Bennett*, G. F. Hewitt*, H. A. Kearsey* and R. K. F. Keeys*

Experiments are described on evaporative heat transfer to boiling water in upflow in a vertical electrically heated 0·497-in inside diameter tube at 1000 lbf/in² (abs.). The main objects were to measure the surface temperature profiles in the region beyond the dry-out point in the channel where liquid ceased to flow on the channel wall, and to investigate the behaviour of the dry-out 'interface' between the 'wetted wall' and the 'dry wall' regions. The test section was made from 'Nimonic' as this can withstand the highest temperatures in the 'dry wall' region and also has a low temperature coefficient of electrical resistivity, thus allowing a uniform heat flux to be maintained with wide axial temperature variation. The temperature in the 'dry wall' region first increased rapidly with distance from the dry-out point, after which it either increased at a slower rate or, at high mass velocities, even decreased. The dry-out 'interface' moved reversibly down and up the channel as the heat flux was increased and decreased. Local surface temperatures showed no hysteresis with cycling of heat flux, in contrast with the pool boiling situation.

A method of predicting the wall temperature profile in the 'dry wall' region has been developed. In this method, the heat-transfer process is considered as being in two steps: wall to superheated steam continuum, and steam continuum to water droplets. The first step was calculated from standard single-phase steam heat-transfer correlations, and the second step was calculated on the basis of simultaneous heat transfer to, and steam diffusion from, the droplets. It was important to take account of the slip between the droplets and the steam. Satisfactory agreement was obtained between measured and predicted wall temperature profiles.

INTRODUCTION

THE 'BURN-OUT' or 'critical heat flux' (CHF) phenomenon, which occurs when the power input to a boiling system exceeds a given value, forms an extremely important limitation on the performance of nuclear boiling-water reactors. The critical heat flux condition (CHFC) corresponds to a termination of good thermal contact between the surface and the liquid phase. Usually, in forced convective (as distinct from pool boiling) systems, CHFC occurs due to a liquid film drying out on the wall. The rest of the liquid is in the form of entrained droplets. We shall term this region of poor thermal contact the 'liquid-deficient region'. Normally, for water, the temperatures in the liquid-deficient region are unacceptably large and a great deal of effort has been expended on the determination of critical heat flux. However, even with reactors which are normally operating in the region where CHFC does not occur, fault conditions can be postulated where a temperature excursion resulting from CHFC is feasible. In understanding such fault conditions, it is necessary to have information on the temperatures attained in the liquid-deficient region. Also, information on heat transfer in the liquid-deficient region is necessary in the design of once-through steam generators of nuclear and non-nuclear types.

Earlier Harwell work on temperatures in the liquid-deficient region (1)† (2) was limited in scope and could not easily be used as a basis for prediction of conditions in a nuclear reactor. It suffered from a further disadvantage that the stainless steel used as a test-section material had a rather high temperature coefficient of electrical resistivity; thus, when the temperature varied widely along the test section, the heat flux also varied widely.

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* *Atomic Energy Research Establishment, Harwell, Didcot, Berkshire.*

† *References are given in Appendix 27.1.*

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This paper describes an extensive series of measurements and an analytical treatment of CHFC heat transfer. The problem of non-uniform heat flux (resulting from resistivity changes in the heated material) has been solved by using tubes of a 'Nimonic' alloy.

A very large number of papers have been written on the CHFC phenomenon. In spite of this, the literature on heat transfer in the liquid-deficient region is rather limited. In water systems, liquid-deficient region temperatures tend to be rather high, and this makes experimentation difficult. This is because the critical heat fluxes (i.e. the minimum heat fluxes at which liquid deficiency can occur) are high for water and the heat-transfer coefficients are low. In cryogenic systems, on the other hand, the critical heat fluxes are very much lower and, in consequence, the temperature rise in the liquid-deficient region is correspondingly smaller. However, cryogenic systems present their own problems in experimentation and the literature in this field is also limited, although some of the most significant papers have appeared in this context. A systematic review of the literature is given elsewhere (**3**).

Notation

A	Cross-sectional area of tube, ft^2.
C_D	Drag coefficient.
C_p	Specific heat of steam at bulk temperature, Btu/lbf.
D	Self-diffusion coefficient, ft^2/h.
D_e	Equivalent or hydraulic mean diameter, ft.
d	Droplet diameter, ft.
d_{CHFC}	Droplet diameter at onset of CHFC, ft.
F	Ventilation factor defined by equation (27.13).
G	Mass velocity, lb/ft^2 h.
g	Acceleration due to gravity, ft/h^2.
h	Heat transfer coefficient, Btu/ft^2 h °F.
h_{TP}	Heat transfer coefficient assuming steam–water mixture to be at saturation temperature, Btu/ft^2 h °F.
k_f	Thermal conductivity of steam phase at film temperature, Btu/ft^2 h °F.
k_G	Thermal conductivity of steam phase, Btu/h ft °F.
L	Distance from start of heated length, ft.
N	Number of droplets flowing per unit time per unit cross-sectional area, h/ft^2.
Pr_f	Prandtl number for steam phase at film temperature.
p	Pressure in bulk steam phase, lbf/ft^2.
p_s	Pressure at steam–water interface, lbf/ft^2.
p_{s1}	Arbitrary chosen pressure close to p_s, lbf/ft^2.
R	Gas constant, Btu/lbf °R.
Re_d	Droplet Reynolds number defined by equation (27.5).
r	Droplet radius ($= d/2$), ft.
T	Temperature in bulk steam phase, °R.
T_s	Temperature at steam–water interface, °R.
T_{s1}	Saturation temperature corresponding to p_{s1}, °R.
t	Time, h.
u_d	Droplet velocity, ft/h.
u_G	Average velocity of steam phase, ft/h.
We	Weber number defined by equation (27.16).
x	Steam quality.
x_{CHFC}	Steam quality at onset of CHFC.
z	Distance from onset of CHFC, ft.
α	Void fraction.
γ	Index for isentropic expansion.
λ	Latent heat of evaporation, Btu/lbf.
μ_f	Viscosity of steam phase at film temperature, lbf/ft h.
μ_G	Viscosity of steam phase, lbf/ft h.
ρ_B	Bulk mean density calculated from equation (27.21), lbf/ft^3.
ρ_G	Steam phase density, lbf/ft^3.
ρ_L	Water phase density, lbf/ft^3.
σ	Surface tension, lbf/ft h.
ϕ	Wall heat flux, Btu/ft^2 h.

EXPERIMENTAL

The present experiments were carried out at a nominal pressure of 1000 lbf/in^2 (abs.). Water was circulated through a preheater and entered the bottom of the test section, which consisted of a vertical 9-ft long, 0·497-in i.d. Nimonic 80A tube. From the test section, the two-phase mixture was returned by way of a condenser to the circulation pump. Twenty-seven thermocouples were attached to the test section at intervals varying from 3 to 12 in. The test section was heated by passing high current, d.c., through its walls; the power supplies, loop, and operating procedure are described in detail in references (**3**), (**4**), and (**5**).

RESULTS AND DISCUSSION

The experimental inside wall temperature profiles and associated inlet temperatures, fluxes, etc. are given in reference (**3**), together with much more extensive graphical presentation.

Temperature profiles

Some 224 temperature profiles were determined during the experiments. Some typical results are shown in Figs 27.1 and 27.2. Each figure shows a set of profiles obtained for constant inlet flow and approximately constant inlet temperature with, in one instance, increasing power and, in the other, decreasing power. In general, for the same inlet conditions and heated lengths, the profiles for increasing and decreasing power are similar, and it will be convenient to discuss the small differences separately below. All the profiles show a characteristic rapid increase in temperature immediately on entering the liquid-deficient region. The thermocouple system used in the present tests would not be able to discriminate the high-frequency temperature changes which are sometimes observed by other experimenters.

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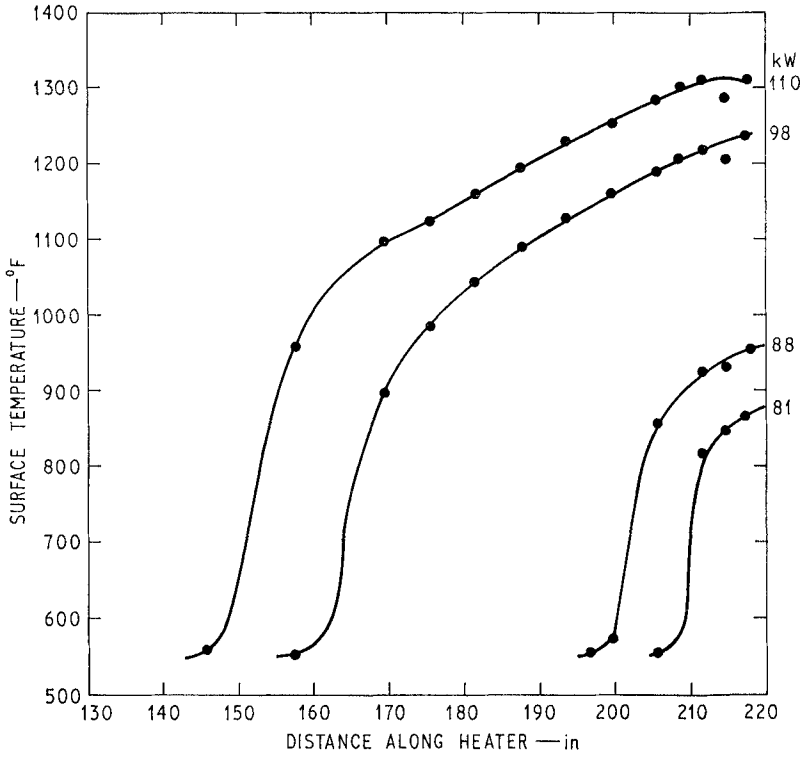


Fig. 27.1. Surface temperature profiles in CHFC for a mass velocity of $0{\cdot}3 \times 10^6$ lbf/ft² h with decreasing heat flux

As the heat flux is increased, the maximum temperature attained also increases and this increase is, of course, maintained in going from the longer heated length to the shorter (Fig. 27.2).

There is a general change in the shape of the temperature profiles in the liquid-deficient region as the mass velocity is increased progressively (compare Figs 27.1 and 27.2). At low mass velocities the temperature continues to increase after the region of initial temperature rise, and rather high temperatures are reached even with low heat fluxes. At medium mass velocities the temperature either rises slowly beyond the initial rise or remains approximately constant. At a mass velocity of $1{\cdot}0 \times 10^6$ (Fig. 27.2) there is already some indication of a temperature drop towards the end of the tube. At the highest mass velocities the temperature passes through a maximum shortly after the point of onset of CHFC. It is also worth noting that increasing the mass velocity restricts the temperature rise and the water phase is obviously playing an important part in the heat-transfer process.

Hysteresis in temperatures at a point along the test section

Fig. 27.3 is a plot of the temperature at a point on the heated surface as a function of heat flux, and shows data in which the liquid-deficient region is entered and left with increasing and decreasing heat flux respectively; there is no hysteresis of the form found in pool boiling. Moreover, there is a small temperature *decrease* on reduction of heat flux, i.e. the opposite of that occurring in pool boiling.

Propagation of the point of onset of CHFC

One of the most important parts of the present test was to establish the reversibility of CHFC and to compare the results obtained in the presence of gross liquid-deficient regions with those obtained in standard tests where CHFC is detected at the end of the uniformly heated channel. The data are presented in terms of CHF versus 'boiling length', as shown in Fig. 27.4. Boiling length is defined as the distance between the point at which the bulk mean enthalpy reaches that of the saturated liquid ($x = 0$) and the point of onset of CHFC. Three types of data are shown:

(1) Standard CHF data [from reference (**6**)], in which inlet subcooling and length have been varied and CHF has been determined by increasing the power until CHFC was detected at *the end of the channel* by means of a 'burn-out detector'.

(2) Data from the present series of tests in which the point of onset of CHFC has been propagated upstream by increasing power.

(3) Data from the present series of tests in which the point of onset of CHFC is moved back downstream by progressively reducing the power.

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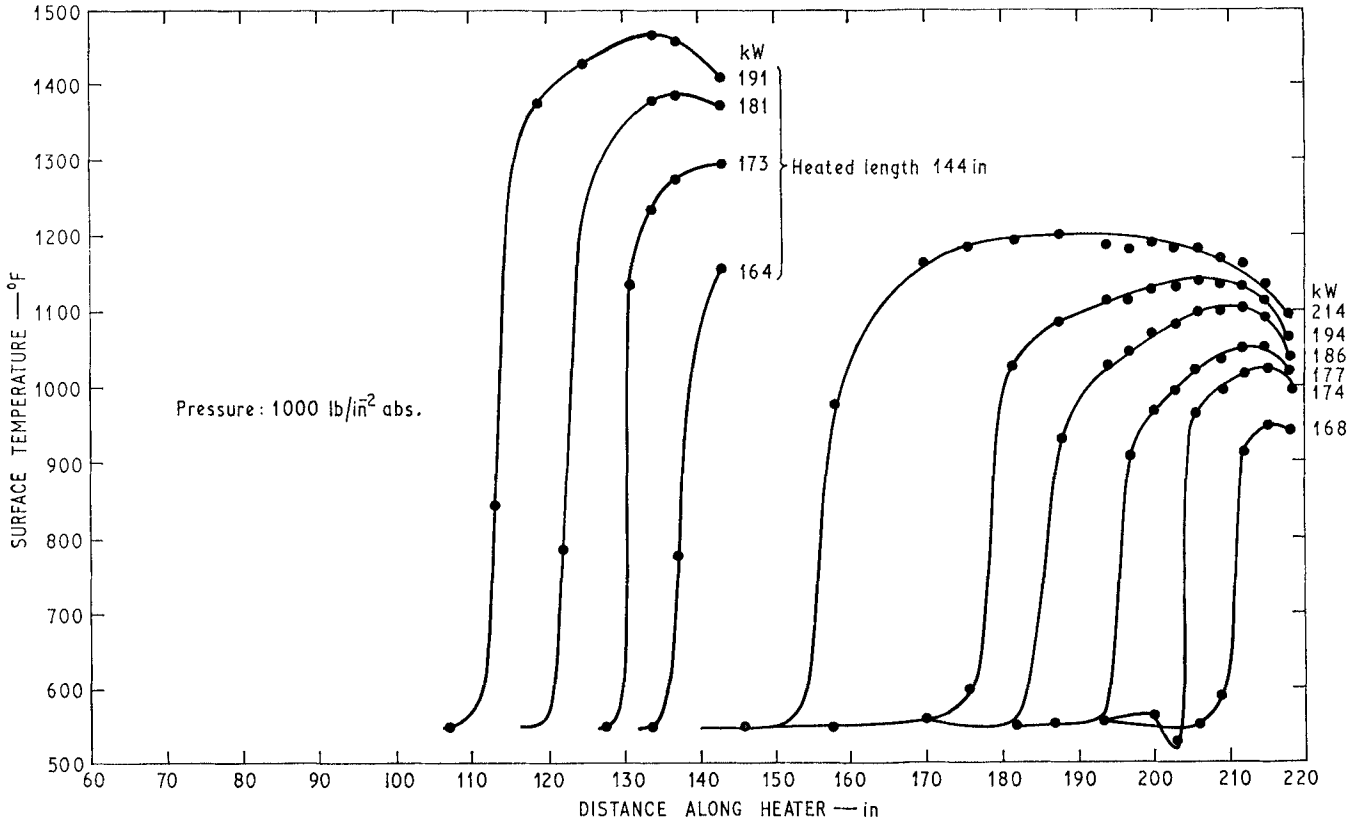
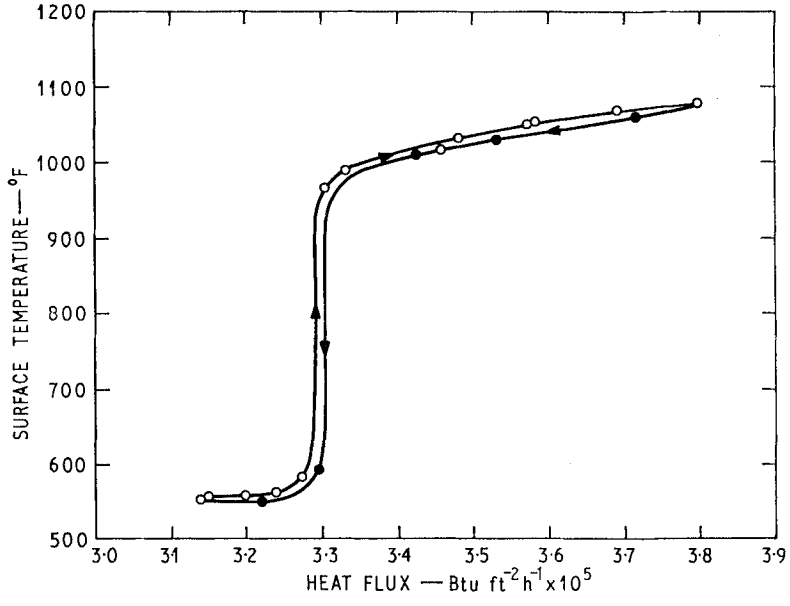


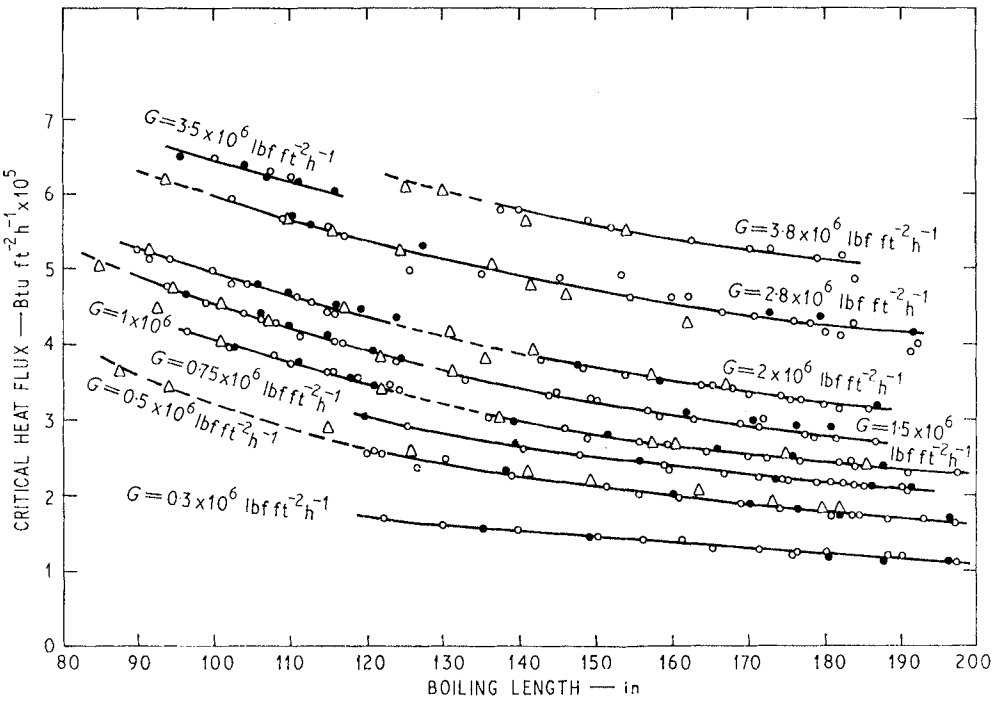
Fig. 27.2. Surface temperature profiles in CHFC for a mass velocity of $1{\cdot}0 \times 10^6$ lbf/ft² h with increasing heat flux



Pressure 1000 lbf/in² (abs.); $G = 1{\cdot}9 \times 10^6$ lbf/ft² h.
○ Flux increasing. ● Flux decreasing.

Fig. 27.3. Temperature at 206 in from start of 219-in heated tube

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Pressure 1000 lbf/in² (abs.); tube diameter 0·497 in.

○ Increasing power. ● Decreasing power. △ CHFC at end of heated length.

Fig. 27.4. Critical heat flux versus boiling length with increasing and decreasing power

For the data from the present series of tests, it was difficult to establish the precise position of the point of onset of CHFC, though by interpolation it could be determined to an accuracy of approximately ±1 in. Data for both the 219-in heated length and the 144-in heated length are shown in Fig. 27.4.

Fig. 27.4 offers convincing proof that the onset of CHFC is reversible and that no excess propagation occurs when gross liquid-deficient regions are present within the channel. Another interesting feature of Fig. 27.4 is that the data for the standard CHF tests agree well with the data obtained here, and this shows that ordinary CHF correlations can be used to predict the *position* of onset of CHFC in uniformly heated channels. It is also interesting to note that the material of construction of the test section did not appear to have a significant effect; the standard CHF determinations (**6**) were carried out with stainless steel tubes, whereas 'Nimonic' tubes were used in the present tests.

PREDICTION OF WALL TEMPERATURE PROFILE

Existing empirical correlations (**7**) (**8**) failed to predict the present data and it did not appear likely that any correlation of a purely empirical type would be applicable. The main difficulty is that the surface temperature is influenced not only by the local vapour and liquid flow rates (which are unknown) but also on the distance from and the conditions at the point of onset of CHFC. Obviously, the real flow situation is exceedingly complex and is well beyond the scope of any analysis which is practicable at the present time. Therefore, in order to develop a usable model, a number of simplifying assumptions must be made.

The basis of the analytical model proposed is that of a two-step heat-transfer process in which heat is transferred from the wall to the flowing vapour and then from the vapour to entrained liquid droplets. The model has some analogies with that proposed by Laverty and Rohsenow (**9**), though here equations are developed which can be integrated along the channel from the point of onset of CHFC.

Basic assumptions of the analysis

The following simplifying assumptions are made; a full discussion of the implications of these assumptions and of their justification is given in reference (**3**).

(1) The droplets do not wet the wall and direct heat transfer from the wall to the droplets can be ignored.

(2) The heat transfer coefficient from wall to fluid is equal to that for steam flowing alone at the same velocity as the average steam velocity in the steam–droplet mixture. For this calculation, the Heineman equation (**10**) was used as follows:

$$\frac{hD_e}{k_f} = 0{\cdot}0157\left(\frac{\rho_G u_G D_e}{\mu_f}\right)^{0{\cdot}84} Pr_f^{1/3}\left(\frac{L}{D_e}\right)^{-0{\cdot}04} \quad \text{for } 6 < L/D < 60 \quad (27.1)$$

$$\frac{hD_e}{k_f} = 0{\cdot}0133\left(\frac{\rho_G u_G D_e}{\mu_f}\right)^{0{\cdot}84} Pr_f^{1/3} \quad \text{for } L/D_e > 60 \quad (27.2)$$

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In applying these equations to the present case, L has been taken as the distance from the point of onset of liquid deficiency.

(3) The interaction between the droplets and the steam can be calculated on the basis of a one-dimensional flow with uniform bulk steam temperature, droplet temperature, and steam and droplet velocity.

(4) The vapour and liquid phases are in thermodynamic equilibrium (i.e. at the saturation temperature) *at the point of onset of CHFC.*

(5) The droplet size is constant at any cross-section of the channel, and no coalescence occurs. Thus, the number of droplets flowing per unit time per unit cross-sectional area is never less than that which occurs at the point of onset of liquid deficiency which is given by the equation:

$$N = \frac{6G(1-x_{\mathrm{CHFC}})}{\pi d^3_{\mathrm{CHFC}}\rho_L} \quad . \quad . \quad (27.3)$$

(6) The relative velocity between the droplets and the vapour is equal, at the point of onset of CHFC, to the free fall velocity of the droplets.

(7) The pressure remains constant in the region considered.

(8) Heat transfer by radiation can be ignored.

Physical processes

The approach made in the present analysis was to set up the basic equations governing the local droplet evaporation, break-up, and acceleration within the channel. These were combined with the mass balance and heat balance equations and with the heat transfer equation [equations (27.1) and (27.2)] and a calculation of wall temperature profile was made by a stepwise integration technique. The acceleration evaporation and break-up of the droplets and the heat and mass balance equations will be considered in separate sections as follows.

Droplet acceleration. As the droplets pass through the channel in the liquid-deficient region, they accelerate continuously owing to increases in the vapour velocity which result from expansion of the vapour due to a rise in temperature and the creation of a new vapour by droplet evaporation. The drag force on the droplet is given by the product $C_D\pi d^2\rho_G(u_G-u_d)^2/8$, where u_D is the droplet velocity, u_G the vapour velocity, and C_D the drag coefficient. Thus, the acceleration of the droplet is given by:

$$\frac{\mathrm{d}u_d}{\mathrm{d}t} = \frac{3C_D\rho_G(u_G-u_d)^2}{4d\rho_L} \quad . \quad . \quad (27.4)$$

The drag coefficient C_D is a function of droplet Reynolds number, defined by the equation:

$$Re_d = \frac{(u_G-u_d)\rho_G d}{\mu_G} \quad . \quad . \quad (27.5)$$

The Ingebo (11) drag law

$$C_D = \frac{27}{Re_d^{0\cdot84}} \quad . \quad . \quad . \quad (27.6)$$

was used. The results were relatively insensitive to the form of drag law employed (3). The rate of change of velocity with distance is thus given by combining equations (27.4) and (27.6) as follows:

$$\frac{\mathrm{d}u_d}{\mathrm{d}z} = \frac{1}{u_d}\frac{\mathrm{d}u_d}{\mathrm{d}t} = \frac{20\cdot25\rho_G^{0\cdot16}\mu_G^{0\cdot84}(u_G-u_d)^{1\cdot16}}{d^{1\cdot84}\rho_L u_d} \quad (27.7)$$

Droplet evaporation. The droplets evaporate as a result of a process of simultaneous heat and mass transfer; heat is transferred to the drop from the surrounding superheated steam, and water vapour diffuses back into the steam from the interface. For the present analysis, a scheme of calculation proposed by Ryley (12) has been employed. For a sphere in a still fluid, the Nusselt number for heat transfer and the Sherwood number for mass transfer are both equal to 2·0. For this case, therefore, the rate of change of droplet radius with time can be given alternatively from the heat transfer or mass transfer equations as follows:

$$\left(\frac{\mathrm{d}r^2}{\mathrm{d}t}\right)_{\mathrm{still}} = \frac{-2k_G(T-T_s)}{\rho_L\lambda} \quad . \quad (27.8)$$

$$\left(\frac{\mathrm{d}r^2}{\mathrm{d}t}\right)_{\mathrm{still}} = \frac{-2D(p_s-p)}{\rho_L RT} \quad . \quad (27.9)$$

Obviously, $T > T_s$ and $p_s > p$ to maintain the heat and diffusive flux respectively. Ryley (12) suggested that the diffusion coefficient should be calculated from the following equation (which is deducible from the kinetic theory of gases):

$$D = \frac{k_G(\gamma-1)}{\rho_G R} \quad . \quad . \quad . \quad (27.10)$$

It is reasonable to suppose that T_s and p_s can be related using saturation data. Over a limited range, the saturation curve can be approximated by the Clausius–Clapeyron equation which can be expressed as follows:

$$\ln\frac{p_s}{p_{s1}} = \frac{-\lambda}{R}\left[\frac{1}{T_s}-\frac{1}{T_{s1}}\right] \quad . \quad . \quad (27.11)$$

Equating the rates of evaporation calculated from equations (27.8) and (27.9) and eliminating T_s by means of equation (27.11), we obtain:

$$T-\frac{1}{\left[\frac{1}{T_{s1}}-\frac{R}{\lambda}\ln\frac{p_s}{p_{s1}}\right]} = \frac{\lambda D}{k_G RT}(p_s-p) \quad (27.12)$$

in which the unknown is p_s. Equation (27.12) may be solved by iterative methods to give p_s and, hence, T_s. Once these values are calculated, they can be substituted into equation (27.8) or equation (27.9) to give the evaporation rate.

Since there is a relative velocity between the droplet and the surrounding gas phase, the actual rate of heat and mass transfer is enhanced. A ventilation factor F can be introduced such that the actual rate of evaporation is given by:

$$\frac{\mathrm{d}r^2}{\mathrm{d}t} = \left(\frac{\mathrm{d}r^2}{\mathrm{d}t}\right)_{\mathrm{still}} \times F \quad . \quad (27.13)$$

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F can be correlated in terms of the droplet Reynolds number defined by equation (27.5) and the vapour Schmidt number. A widely used form of this correlation is as follows:

$$F = 1+0{\cdot}276Re_d^{1/2}\left(\frac{\mu_G}{D\rho_G}\right)^{1/3} \qquad (27.14)$$

The stages for the calculation of evaporation rate are then as follows:

(*a*) For the local gas physical properties calculate D, and for the local values of T and p calculate p_s, from equation (27.12).

(*b*) From the calculated value of p_s, evaluate T_s and use equation (27.8) to calculate the evaporation rate for the stationary drop.

(*c*) Using the local droplet Reynolds number, calculate F from equation (27.14) and determine the evaporation rate using this value of F in equation (27.13).

The rate of change of droplet diameter with length is then given by the relationship:

$$\frac{dd}{dz} = \frac{1}{u_d}\left[\frac{2}{d}\frac{dr^2}{dt}\right] \qquad (27.15)$$

Droplet break-up. Since the slip between the droplets and the gas tends to increase along the channel, a point may be reached at which the droplets divide into two. Following Forslund and Rohsenow (**13**) this effect has now been allowed for in the analysis described here; a critical Weber number is defined by the equation [see, however, discussion in reference (**3**)]:

$$We = \frac{G(u_G-u_d)^2 d}{\sigma g} = 7{\cdot}5 \qquad (27.16)$$

In practice, however, the droplet Weber numbers for the optimum droplet size did not usually reach the value of 7·5 suggested by Forslund and Rohsenow.

Heat and mass balances. The rate of change of steam quality with length is given by the mass balance equation:

$$\frac{dx}{dz} = \frac{-N\pi\rho_L d^2}{2G}\frac{dd}{dz} \qquad (27.17)$$

Similarly, the rate of change of steam temperature (T) with length can be determined by means of a heat balance as follows:

$$\frac{dT}{dz} = \frac{\phi\pi D - AG\lambda(dx/dz)}{AG.C_p} \qquad (27.18)$$

Solution of equations

Equations (27.7), (27.15), (27.17), and (27.18) are four differential equations which must be satisfied simultaneously as the flow proceeds along the channel. These four differential equations were solved using an integration method of the Runge–Kutta type. The integration was stepped forward from the point of onset of CHFC upstream along the channel and the wall temperature calculated at each point. The starting value of droplet size was varied until the best fit was found. It was noted (**3**) that the predicted temperature profile varies only relatively slowly with the droplet size chosen. Great sensitivity to droplet size would have made the present method of analysis unsuitable. Fig. 27.5 shows the effect of quality and mass velocity on the droplet size for optimum fit. The choice of a constant droplet size of 0·3 mm for the whole range would give a moderately good fit, with the exception of the high mass velocity region. Even in this region, however, the use of a constant droplet size still leads to predictions which are better than other available methods.

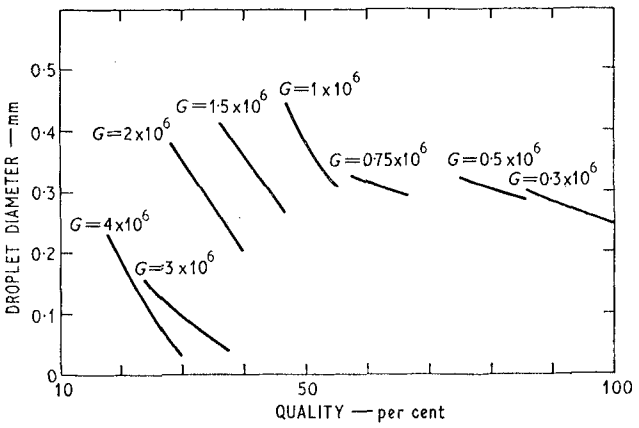


Fig. 27.5. Variation of droplet diameter with quality and mass velocity

It is of interest to compare the method proposed here with existing correlations and calculation techniques, and such comparisons are given in Fig. 27.6. In addition to the experimental results and the curve calculated using the above analytical method, the following curves are also shown:

(1) A curve calculated assuming that, from the point of onset of CHFC, no further evaporation takes place and that the transfer is to the steam phase alone, continuing to flow at its mass velocity at the onset point and progressively increasing its temperature as it proceeds along the channel. In other words, the water phase is disregarded.

(2) A line calculated assuming that the evaporation of the water droplets proceeds rapidly enough to maintain thermodynamic equilibrium within the steam–droplet mixture. Thus, the steam temperature remains at the saturation value until the condition of 100 per cent thermodynamic quality is reached, after which point the steam begins to be superheated. With this assumption, the wall temperature falls initially (owing to the increase in steam velocity) and then, if the condition of 100 per cent thermodynamic quality is reached in the channel, the wall temperature begins to rise again as the steam becomes superheated.

(3) A line calculated from the GE (**8**) correlation:

$$\frac{h_{TP}D_e}{(Pr_f)^{1/3}k_f}\left(\frac{1-\alpha}{\alpha}\right) = 0{\cdot}00136\left[\frac{D_eG}{\mu_f}\left(\frac{1-x}{x}\right)\right]^{0{\cdot}853} \qquad (27.19)$$

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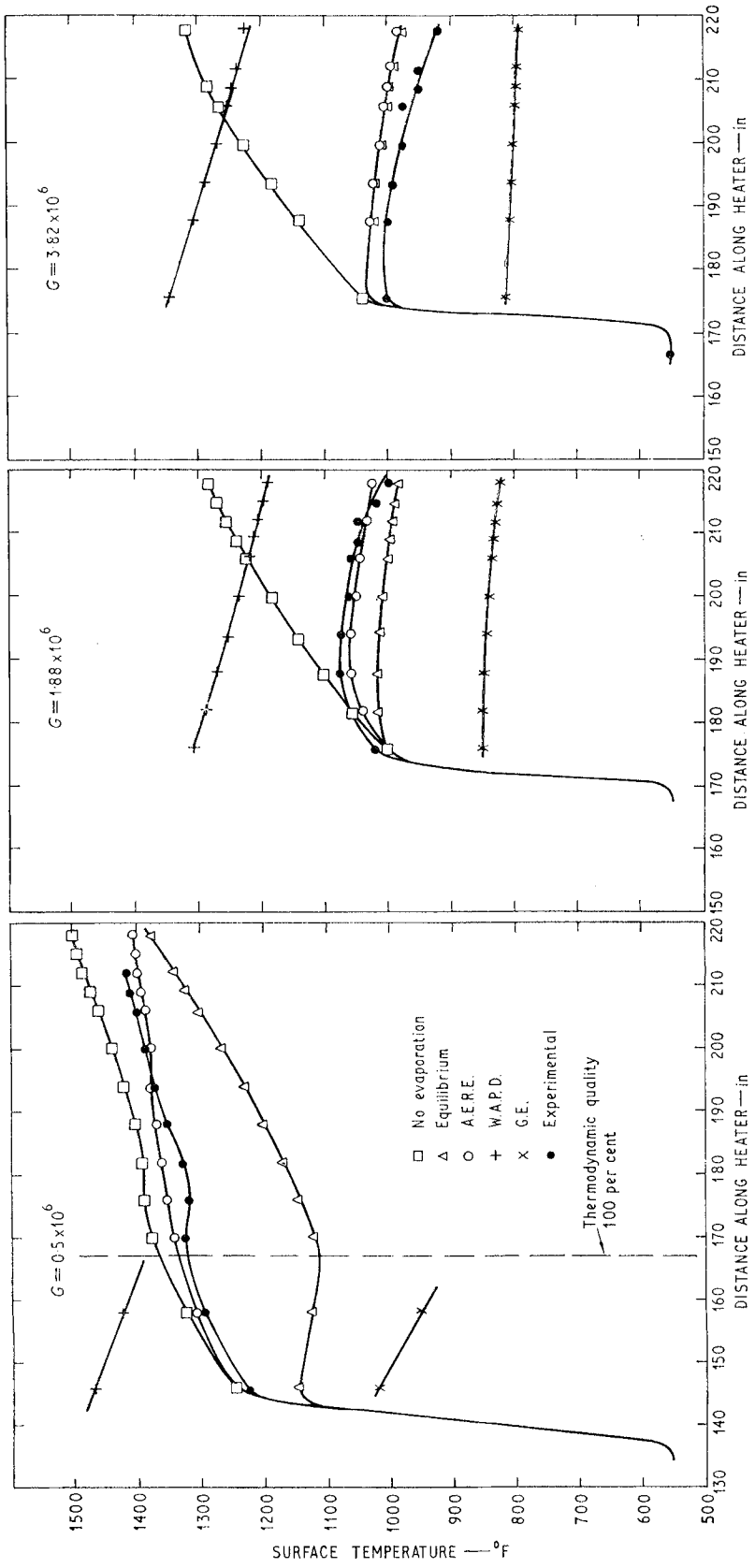


Fig. 27.6. Comparison of calculation methods and experiment

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where the heat-transfer coefficient h_{TP} is calculated assuming the steam–water mixture to be at saturation temperature, D_e is the hydraulic diameter, G the mass velocity, and Pr_f and μ_f are the Prandtl number, thermal conductivity, and viscosity of the steam phase taken at a temperature which is the arithmetic mean of the heater wall and bulk (saturation) temperature. x is the steam quality and α the void fraction which is calculated from the expression:

$$\alpha = \frac{1}{1+\frac{(1-x)}{x}\left(\frac{\rho_G}{\rho_L}\right)^{2/3}} \quad . \quad . \quad (27.20)$$

where ρ_G and ρ_L are the vapour and liquid densities respectively.

(4) A line calculated from the WAPD (7) correlation:

$$\left(\frac{h_{TP}D_e}{k_f}\right) = 0{\cdot}0193\left(\frac{D_eG}{\mu_f}\right)^{0{\cdot}80} Pr_f^{1{\cdot}23}\left(\frac{\rho_G}{\rho_B}\right)^{0{\cdot}68}\left(\frac{\rho_G}{\rho_L}\right)^{0{\cdot}068} \quad . \quad . \quad . \quad (27.21)$$

where ρ_B is the bulk mean density assumed to be given by the homogeneous model:

$$\rho_B = \frac{1}{x/\rho_G+(1-x)/\rho_L} \quad . \quad . \quad (27.22)$$

In equation (27.22) the quality x is calculated on a thermodynamic equilibrium basis.

It will be seen from Fig. 27.6 that the GE correlation consistently underpredicts the wall temperatures, whereas the WAPD correlation overpredicts the temperature. Another general observation is that the temperature profile approaches that for thermodynamic equilibrium for high mass velocities and approaches that for no evaporation at low mass velocities. The present results are thus consistent with the earlier work carried out at low mass velocity at Harwell (2). The analytical model proposed here is successful in predicting this change of behaviour with increasing mass velocity and, at the highest mass velocities (see Fig. 27.6), there is practically no difference between the wall temperatures predicted from the model and those predicted assuming thermodynamic equilibrium. It is interesting to note in Fig. 27.6 that the wall temperatures are slightly lower than those predicted for thermodynamic equilibrium, and this may suggest that some enhancement of heat transfer is occurring, for example, as a result of drop–wall interaction. However, it should be pointed out that the discrepancy between the experimental and measured temperatures is rather small and could be accounted for in the error-band around the correlation used for calculation of heat transfer to steam.

The analytical method described above has been applied to the high-pressure data of Bishop *et al.* (7). Although the analysis predicts the peak temperatures reasonably accurately, the predicted decrease in temperature in the CHFC region is much slower than that observed experimentally. This may have been caused by mechanisms that were neglected in the present analysis. It should be borne in mind that the temperature peaks of the type obtained by Bishop *et al.* also occur in the supercritical region where, of course, the concept of a dispersed droplet flow can no longer be valid. It seems possible, therefore, that there is a gradual change of mechanism as the pressure is increased. This change may take the form of an approach to a film-boiling type of mechanism in which a vapour or vapour-like layer near the surface provides an insulation between the surface and the bulk liquid or liquid-like fluid in the centre of the channel.

CONCLUSIONS

The following main conclusions can be drawn.

(1) Measurements carried out over a wide range of heat fluxes, lengths, and mass velocities show that the temperature increases sharply at the point of onset of CHFC and then increases less sharply from this point onwards. As the heat flux is increased, the maximum temperature in the tube is increased and the point of onset is propagated systematically upstream.

(2) The shape of the wall temperature versus distance curve in the liquid-deficient region is dependent on the mass velocity. At low mass velocity the temperature increases continuously, whereas at high mass velocity the temperature passes through a maximum value.

(3) At a given point on the tube wall, the temperature rises sharply as CHFC is reached and then less sharply with further increases of flux. When the flux is *reduced*, the temperature–flux curve is close to that observed for increasing flux. This reversibility in temperature contrasts markedly with electrically heated pool-boiling systems.

(4) The point of onset of CHFC is propagated reversibly as flux is increased and then decreased.

(5) The position of the point of onset of CHFC agrees well with that expected from measurements of CHF by detection of temperature rise at the end of channels of various lengths with various inlet subcoolings.

(6) Comparison of the liquid-deficient region heat-transfer data with existing empirical correlations showed gross differences, and we conclude that purely empirical methods of correlation are unlikely to be satisfactory for this region.

(7) Comparison of the wall temperature data in the liquid-deficient region with that calculated assuming no evaporation and complete thermodynamic equilibrium respectively showed that the former condition was approached at low mass velocity and the latter at high mass velocity.

(8) An analytical model has been developed which is successful in predicting the wall temperature profiles providing an initial droplet size is assumed. However, the results were not particularly sensitive to the choice of initial droplet size. The analysis quite accurately predicts the change in shape of the wall temperature–distance curve with mass velocity.

(9) In principle, the analysis can be extended to much wider ranges of fluid conditions than those found in the present tests. However, comparison of the analysis with

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data at high pressures (approaching the critical point) seems to indicate that other mechanisms are becoming important. Also, the analysis may be inapplicable at very low qualities (and certainly in the subcooled region) where the mechanism proposed in the analysis can no longer apply.

APPENDIX 27.1

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